Mehrnoosh Hasanzade

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No sponsorship required

PROFESSIONAL SUMMARY

Senior Data Scientist with a PhD in Economics and experience in predictive modeling, machine learning, causal inference, experimentation, and applied AI. Skilled in Python, SQL, statistical modeling, LLM evaluation, and retrieval-augmented generation (RAG). Experienced in analyzing structured and unstructured data to deliver actionable business and research insights.

SKILLS

• **Programming & Data:** Python, SQL, Pandas, NumPy, PostgreSQL, BigQuery, Snowflake, Git
• **Machine Learning & Statistics:** Scikit-learn, XGBoost, PyTorch, TensorFlow, Classification, Regression, Clustering, Segmentation, Feature Engineering, Cross-Validation, Model Evaluation, Bayesian Inference, Time-Series Modeling
• **LLMs & NLP:** LLM Applications, Retrieval-Augmented Generation (RAG), AI Agents, Tool/Function Calling, Prompt Engineering, LLM Evaluation, Embeddings, Vector Search, FAISS, Transformers, Text Classification, Information Extraction
• **Experimentation & Causal Inference:** A/B Testing, Hypothesis Testing, Uplift Modeling, Experiment Validation
• **Tools & Platforms:** AWS, REST APIs, FastAPI, Docker, Tableau, Looker, Gradio

PROFESSIONAL EXPERIENCE

Senior Data Scientist, Ovative Group | May 2026 – Jul 2026
• Developed and evaluated Bayesian marketing mix models using Python to assess channel performance, validate model reliability, and support data-driven investment decisions.
• Collaborated with product teams, data scientists, and business stakeholders to translate marketing analytics requirements into statistical modeling solutions and actionable recommendations.

Research Data Scientist, Annenberg Social Action Lab, University of Pennsylvania | Sep 2024 – Dec 2025
• Developed an LLM-based pipeline to identify and classify public health misinformation in social media, generating insights into high-risk narratives to inform research and policy.
• Conducted LLM evaluation against gold-standard human annotations, using error analysis, prompt refinement, and expanded labeled data to improve classification performance.
• Built a clinical RAG system using hierarchical chunking, MiniLM embeddings, FAISS, GPT-4o, and Gradio to enable grounded question answering over clinical documents.
• Analyzed large-scale survey data using demographic and behavioral segmentation to identify HIV/STI awareness trends; led cross-functional research and communicated findings to NIH stakeholders.

Data Analyst, Governance & Quality Management (Contract), Mozilla | Jul 2023 – Dec 2023
• Built Python and SQL workflows in BigQuery and Zendesk environments to clean, transform, and standardize multi-source customer and operational data for analytics and reporting.
• Validated customer experience metrics through QA, reconciliation, and industry benchmarking, reducing data-related errors by 10%.
• Developed Looker dashboards to surface KPIs, trends, and operational insights for business stakeholders.

Data Scientist, Constellation Agency LLC | Jun 2022 – Jun 2023
• Built Python and PyTorch data pipelines integrating structured data and unstructured text to support machine learning and healthcare analytics.
• Developed Python and Snowflake SQL workflows integrating patient, claims, clinical, and interaction data to support healthcare provider segmentation and analysis.
• Applied machine learning and statistical modeling to evaluate pharmaceutical campaign performance, supporting marketing mix modeling, attribution, and investment decisions.
• Designed and analyzed A/B tests and uplift modeling frameworks to estimate the causal impact of marketing tactics and guide targeting decisions; communicated findings through Tableau and Looker dashboards.

SELECTED PROJECTS

Agentic AI Customer Support System | Python, Qwen3, Ollama, SQLite, LLM Tool Calling GitHub
• Developed a text-based LLM support agent to handle customer and billing inquiries, integrating Qwen3, SQLite-backed Python tools, and refund confirmation workflows to demonstrate multi-step tool execution and refund validation and duplicate-refund safeguards.

Campaign Optimization & Sales Prediction | Python, XGBoost, Scikit-learn | GitHub
• Developed and tuned an XGBoost predictive model for customer purchase propensity, achieving 0.732 test ROC-AUC and 71.3% precision among the top 10% of ranked customers on synthetic data.

A/B Testing & Retention Analysis | Python, Statistical Testing | GitHub
• Evaluated a mobile-game retention experiment using hypothesis testing, Holm correction, and bootstrap confidence intervals, identifying 0.82 percentage points lower Day-7 retention in the treatment group and flagging sample-ratio mismatch as a validity risk.

Clinical Notes NLP & Agentic RAG | Python, Sentence Transformers, FAISS, GPT-4o-mini GitHub
• Built a pipeline over synthetic clinical notes to extract sections, lab values, demographic features, and negated findings, then combine semantic search with patient-level summaries through a three-tool retrieval agent.

EDUCATION

Doctor of Philosophy in Economics, University of Wisconsin–Milwaukee | 2017–2022
Master of Project Management (MPM), Tennessee Wesleyan University | 2024–2025
Master of Science in Economics, Sharif University of Technology | 2012–2014
Bachelor of Science in Electrical Engineering, University of Birjand | 2004–2009